ACTL3182 Wk8

Tadhg XG

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Revision: Brownian Motion

Brownian Motion

A stochastic process $\{W_t : t \geq 0\}$ is said to be a $\mathbb{P}$-Brownian motion if

1. W_t has continuous sample paths and $W_0 = 0$;
2. Under $\mathbb{P}$, $W_t \sim \mathcal{N}(0, t)$, $t \geq 0$;
3. Under $\mathbb{P}$, $W_{s+t} - W_s \sim \mathcal{N}(0, t)$ and is independent of $\mathcal{F}_s$, where $\mathcal{F}_s$ is the history of W_t up until time s, $s, t \geq 0$.

Revision: Itô Lemma

Itô lemma

Let $f(t, x)$ be a function and X_t an Itô process. Then,

$$\mathrm{d}f(t, X_t) = f_t(t, X_t)\,\mathrm{d}t + f_x(t, X_t)\,\mathrm{d}X_t + \frac{1}{2} f_{xx}(t, X_t)\,(\mathrm{d}X_t)^2$$

Combined with the heuristics

$$(\mathrm{d}t)^2 = 0, \quad (\mathrm{d}t)(\mathrm{d}W_t) = 0, \quad (\mathrm{d}W_t)^2 = \mathrm{d}t,$$

this can be used to find dynamics of functions of Itô processes.

CMG Theorem

Theorem (CMG)

Let W_t be a $\mathbb{P}$-Brownian motion and γ_t be another stochastic process. Then, there exists a measure $\mathbb{Q}$ such that the process

$$W_{\mathbb{Q}}(t) = W(t) + \int_0^t \gamma(s)\,\mathrm{d}s$$

is a $\mathbb{Q}$-brownian motion.

We can rewrite the expression for $W_{\mathbb{Q}}$ as

$$\mathrm{d}W_{\mathbb{Q}}(t) = \mathrm{d}W(t) + \gamma(t)\,\mathrm{d}t, \qquad W_{\mathbb{Q}}(0) = 0.$$

Itô Integral as a Martingale

Recall the Itô integral

$$I(t) = \int_0^t \sigma(s)\,\mathrm{d}W(s),$$

where $W(\cdot)$ is a $\mathbb{P}$-Brownian motion. Then, it can be shown that $I(t)$ is a $\mathbb{P}$-Martingale.

So, if a process X_t has dynamics

$$X_t = \sigma(t)\,\mathrm{d}W_t,$$

then it must be a $\mathbb{P}$-martingale.

Combining Itô Integral and CMG

We want to find a measure $\mathbb{Q}$ such that the discounted stock price $Z(t)$ is a $\mathbb{Q}$-martingale, for this will give us an explicit pricing formula (as seen in the binomial tree model).

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We want to find a measure $\mathbb{Q}$ such that the discounted stock price $Z(t)$ is a $\mathbb{Q}$-martingale, for this will give us an explicit pricing formula (as seen in the binomial tree model).

Say I know that $Z(t)$ has the dynamics under $\mathbb{P}$

$$\mathrm{d}Z(t) = \mu(t)\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W(t).$$

Then, by CMG theorem, for any process $\gamma(t)$ there exists a measure $\mathbb{Q}$ such that the process satisfying

$$\mathrm{d}W_{\mathbb{Q}}(t) = \mathrm{d}W(t) + \gamma(t)\,\mathrm{d}t$$

is a $\mathbb{Q}$-Brownian motion...

Market Price of Risk

So,

$$\begin{aligned} \mathrm{d}Z(t) &= \mu(t)\,\mathrm{d}t + \sigma(t)(\mathrm{d}W_{\mathbb{Q}}(t) - \gamma(t)\,\mathrm{d}t) \\ &= (\mu(t) - \sigma(t)\gamma(t))\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_{\mathbb{Q}}(t). \end{aligned}$$

Choose $\gamma(t) = \mu(t)/\sigma(t)$ so that the $\mathrm{d}t$ term disappears, and then

$$\mathrm{d}Z(t) = \sigma(t)\,\mathrm{d}W_{\mathbb{Q}}(t).$$

So $Z(t)$ is a $\mathbb{Q}$-martingale!

The choice of $\gamma(t)$ that gives us such $\mathbb{Q}$ is called the market price of risk.